

DAY 4

Scenario Planning and Sensitivity Analysis

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

Lesson Forge Pro (owner)

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This chapter builds working competence in Scenario Planning and Sensitivity Analysis: the concepts, the standard procedure, and the judgment to apply it under exam conditions.

Concepts and vocabulary

Scenario planning and sensitivity analysis share a common purpose: they help analysts understand how uncertainty affects a model's conclusions before that uncertainty becomes reality. Mastering the vocabulary is the essential first step, because every conversation with a client, colleague, or examiner depends on using these terms precisely.

Core Definitions

A model is a simplified mathematical representation of a real-world situation—a set of inputs, relationships, and outputs that lets you ask structured questions. In financial modeling, the output is typically a value metric such as net present value (NPV), earnings per share, or free cash flow.

An input variable (also called a driver or assumption) is any number you feed into the model. Examples include revenue growth rate, discount rate, or unit cost. Notation: inputs are commonly labeled $x_1, x_2, \dots, x_n$.

An output variable (the response or target) is what the model calculates. Notation: $Y = f(x_1, x_2, \dots, x_n)$.

Scenario planning asks: "What if the world looks like this?" You define a coherent combination of input values—a scenario—and observe how Y changes. The classic trio is a base case (most likely), upside/best case, and downside/worst case. Each scenario is internally consistent; you don't mix a pessimistic revenue assumption with an optimistic cost assumption unless there is a logical reason to do so.

Sensitivity analysis asks a narrower question: "How much does Y change when I move one input by a defined amount, holding everything else constant?" This one-at-a-time approach isolates the marginal impact of each driver and ranks them by influence. The core measure is the sensitivity coefficient:

$$S_i = \frac{\partial Y}{\partial x_i}$$

or, in percentage terms (useful for comparing drivers with different units):

$$S_i (\%) = \left(\frac{\partial Y}{\partial x_i} \cdot \frac{x_{i, \text{base}}}{Y_{\text{base}}} \right) \times 100$$

A tornado chart visualizes sensitivity coefficients ranked from largest to smallest impact, giving the diagram its characteristic funnel shape.

Where It Fits in Practice

Both techniques live inside the broader discipline of financial planning and analysis (FP&A) and management decision-making. They answer three classic questions: "What is?" (base case), "What if?" (scenario and sensitivity), and "What has to be done?" (goal-seeking and reverse engineering). You will encounter them in project valuation via discounted cash flow, capital budgeting, revenue forecasting, and M&A transaction analytics—anywhere a decision depends on uncertain assumptions.

Worked Example: Terminology Applied to a Mini Scenario

A small retailer is evaluating a new product line. The analyst builds a one-year profit model:

$$\text{Profit (Y)} = (\text{Price} \times \text{Units}) - \text{Fixed Costs}$$

Base-case inputs:

- Price (x_p) = \$50 per unit
- Units sold (x_u) = 2,000
- Fixed costs (x_f) = \$60,000

Step 1 — Calculate base-case output.

$$Y_{\text{base}} = (\$50 \times 2,000) - \$60,000 = \$100,000 - \$60,000 = \$40,000$$

Step 2 — Define a downside scenario. Suppose a competitor enters the market: price falls to \$44 and units drop to 1,700 (a coherent combination).

$$Y_{\text{downside}} = (\$44 \times 1,700) - \$60,000 = \$74,800 - \$60,000 = \$14,800$$

The scenario output is \$14,800—a 63% decline from the base case.

Step 3 — Run a sensitivity analysis on price alone. Hold units and fixed costs at base-case values; reduce price by \$5 (a 10% decrease).

$$S_{\text{price}} = \frac{(\$10,000 - \$40,000) / (\$5 - \$50)}{(\$5 - \$50) / \$50} = \frac{(-\$30,000) / (-\$45)}{(\$45) / \$50} = \frac{666.67}{90} = 7.41$$

$$S_{\text{price}} (\%) = \left(\frac{\$10,000 - \$40,000}{\$40,000} \right) / \left(\frac{\$5 - \$50}{\$50} \right) \times 100 = \left(\frac{-25\%}{-10\%} \right) \times 100 = 250\%$$

A 1% change in price produces a 2.5% change in profit—price is a highly leveraged driver.

Step 4 — Interpret. The sensitivity coefficient of 2.5 tells management that pricing discipline is critical; the downside scenario quantifies what a realistic competitive threat actually costs in dollar terms.

Common mistake: Confusing a scenario with a sensitivity test. A scenario changes multiple inputs simultaneously in a logically consistent story. A sensitivity test changes one input at a time to measure its isolated effect. Mixing the two—adjusting several variables and calling it a sensitivity analysis—obscures which driver is actually responsible for the output change.

With these building blocks in place—inputs, outputs, the $Y = f(x)$ notation, scenario coherence, and the sensitivity coefficient—you have the vocabulary to read any scenario or sensitivity table and immediately understand what question it is answering.

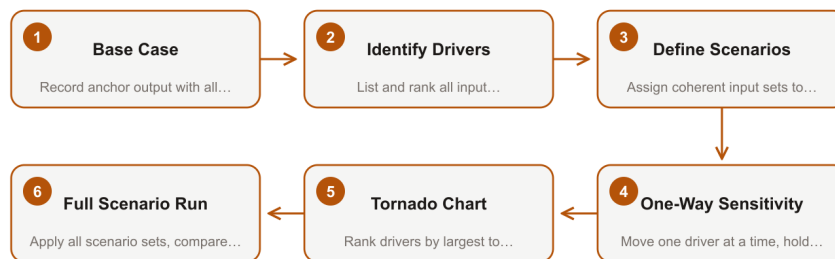
The standard procedure

Scenario planning and sensitivity analysis share a common backbone: a structured, repeatable procedure that moves from clearly defined inputs through a model to outputs you can interrogate and trust. Mastering that procedure is what separates analysts who produce actionable insight from those

who produce impressive-looking spreadsheets nobody acts on.

Inputs

Standard Scenario & Sensitivity Procedure



Six repeatable steps from assumption gathering to actionable scenario output

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Before touching a formula, gather and document four categories of inputs:

- Base-case assumptions — the single "most likely" value for each driver (e.g., revenue growth rate, operating margin, discount rate).
- Driver ranges — realistic low and high bounds for each assumption, grounded in historical data, industry benchmarks, or expert judgment.
- The financial model — a working model that maps inputs to outputs through explicit, auditable formulas.
- Output metric(s) — the decision-relevant result you will track, such as Net Present Value (NPV), EBITDA, or free cash flow.

The Step-by-Step Procedure

Step 1 — Establish the base case.

Run the model with all base-case assumptions and record the output. This is your anchor.

Step 2 — Identify key drivers.

List every assumption that feeds the output. Rank them by perceived importance; you will test all of them but focus narrative attention on the top three to five.

Step 3 — Define scenarios.

Create at least three named scenarios: Base, Upside, and Downside. For each scenario, assign a coherent set of values to every driver. Coherence matters — a downside scenario should not pair a recession-level revenue drop with a boom-level margin expansion.

Step 4 — Run one-way sensitivity analysis.

Hold all drivers at base-case values. Move a single driver to its low value, record the output change, reset it, then move it to its high value. Repeat for every driver. This isolates each driver's individual impact.

Step 5 — Rank drivers by impact (tornado chart).

Sort drivers from largest to smallest swing in the output metric. The widest bar on a tornado chart is your most sensitive assumption — it deserves the most scrutiny.

Step 6 — Run full scenario analysis.

Apply each complete scenario (all drivers shifted simultaneously) and record the output for each.

Step 7 — Document and present.

Summarize results in a sensitivity table and scenario summary table. Communicate the range of outcomes and the key drivers of that range.

Worked Example

A project analyst is evaluating a two-year project. The base-case NPV calculation uses:

Driver	Downside	Base	Upside
Annual revenue (\$M)	8.0	10.0	12.0
Operating margin (%)	18%	22%	26%
Discount rate (%)	12%	10%	8%

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Base-case NPV calculation (simplified, two equal annual cash flows):

$$\text{NPV} = \frac{\text{Revenue} \times \text{Margin}}{\text{Initial Investment} \times (1+r)} + \frac{\text{Revenue} \times \text{Margin}}{\text{Initial Investment} \times (1+r)^2}$$

Initial investment = \$3.0 M. Annual operating cash flow = \$10.0 M × 22% = \$2.20 M.

$$\text{NPV} = \frac{2.20}{1.10} + \frac{2.20}{1.10^2} \div 3.0 = \frac{2.000}{3.0} + \frac{1.818}{3.0} = \$0.818 \text{ M}$$

One-way sensitivity — revenue driver:

- Low revenue: \$8.0 M × 22% = \$1.76 M/yr !' NPV = 1.600 + 1.455 " 3.0 = \$0.055 M
- High revenue: \$12.0 M × 22% = \$2.64 M/yr !' NPV = 2.400 + 2.182 " 3.0 = \$1.582 M
- Swing = \$1.582 " \$0.055 = \$1.527 M

One-way sensitivity — discount rate driver (revenue and margin held at base):

- High rate (12%): NPV = 2.20/1.12 + 2.20/1.12² " 3.0 = 1.964 + 1.754 " 3.0 = \$0.718 M
- Low rate (8%): NPV = 2.20/1.08 + 2.20/1.08² " 3.0 = 2.037 + 1.886 " 3.0 = \$0.923 M
- Swing = \$0.205 M

Revenue produces a swing nearly 7.5× larger than the discount rate, so it tops the tornado chart and demands the most due diligence.

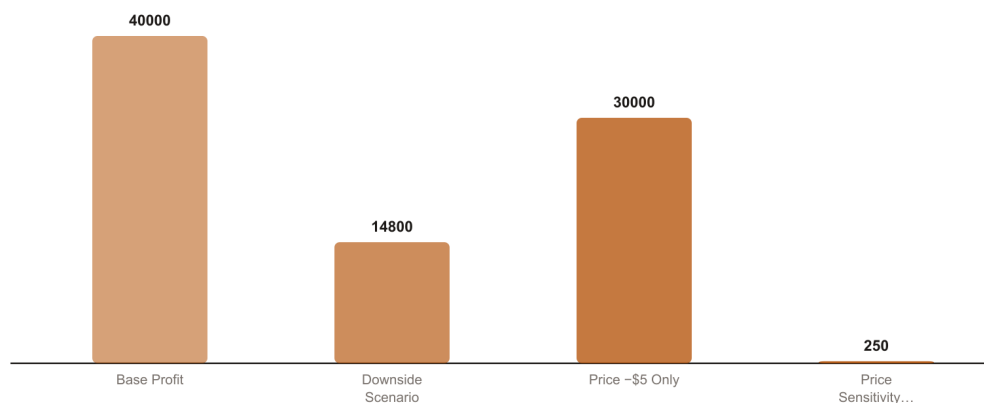
Full downside scenario (all drivers at worst simultaneously):

Annual cash flow = $\$8.0 \text{ M} \times 18\% = \1.44 M !' NPV = $1.44/1.12 + 1.44/1.12^2 + \dots + 1.44/1.12^{3.0} = 1.286 + 1.148 + \dots = \0.566 M

The project destroys value under a coherent downside — a critical finding for the decision-maker.

Outputs and Checks

Sensitivity: Price vs. Profit Impact



Downside scenario cuts profit 63%; price sensitivity coefficient of 2.5 flags pricing as top risk

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Your deliverables are: a base-case output, a sensitivity (tornado) table, and a scenario summary table showing NPV (or your chosen metric) under each named scenario.

Checks to run before presenting:

- Verify that changing a driver in isolation moves the output in the expected direction (sign check).
- Confirm scenarios are internally consistent — no contradictory driver combinations.
- Ensure the base case sits between downside and upside outputs; if it does not, a formula error is likely.

Common mistake: Analysts often run sensitivity analysis on every input with equal weight, then bury the audience in a 20-row tornado chart. Focus your narrative on the top three to five drivers. More rows do not mean more insight — they mean more noise.

Applications and pitfalls

Scenario planning and sensitivity analysis appear in virtually every corner of applied finance—from a startup pitching investors to a multinational stress-testing its balance sheet. Mastering the mechanics is only half the battle; knowing where to apply them and what can go wrong separates competent analysts from excellent ones.

Typical Use Cases

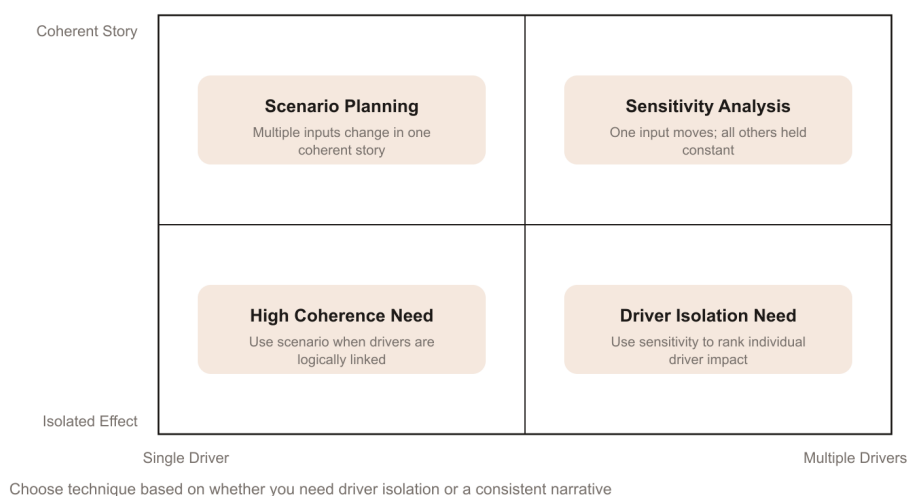
The most common applications cluster around decisions where the future is genuinely uncertain and the stakes are high enough to justify the modeling effort:

- Business and project valuation (DCF): Analysts build a base case, then flex key drivers—revenue growth, operating margins, discount rate—to bound the range of plausible values.
- FP&A and budgeting: Finance teams answer "what if" questions (what if raw-material costs rise 15%?) and "what has to be done" questions (what revenue growth is required to hit the EBITDA target?).
- Capital budgeting and WACC analysis: Changing the cost of equity or the debt/equity mix can swing an NPV from positive to negative; sensitivity tables make this visible to decision-makers.
- Credit analysis and impairment modeling: Lenders stress-test debt-service coverage under adverse scenarios to set covenant thresholds.
- M&A and LBO modeling: Deal teams examine how entry multiple, exit multiple, and leverage interact to determine equity returns.

In every case the goal is the same: replace a false sense of precision with a structured picture of uncertainty.

Common Mistakes

Scenario vs. Sensitivity: When to Use



Choose technique based on whether you need driver isolation or a consistent narrative

Varying one input while ignoring correlated inputs. Revenue and cost of goods sold are not independent. If you flex revenue upward without adjusting variable costs, your margin expands unrealistically. Always ask whether inputs move together in the real world.

Choosing arbitrary shock sizes. Shifting every input by $\pm 10\%$ is convenient but rarely meaningful. Use historically observed volatility, management guidance ranges, or contractual bounds to calibrate shocks.

Confusing scenario analysis with sensitivity analysis. Sensitivity analysis changes one variable at a time to measure its marginal impact. Scenario analysis changes multiple variables simultaneously to represent a coherent story (e.g., a recession scenario). Mixing the two produces outputs that are hard to interpret.

Anchoring on the base case. Analysts sometimes build elaborate downside and upside cases but never seriously question whether the base case itself is realistic. The base case should be stress-tested just like the alternatives.

Common mistake: Building a two-variable data table and reading the diagonal as "the most likely path." Each cell in a sensitivity table is an independent what-if, not a sequence of events. The diagonal has no special probabilistic meaning unless you have explicitly modeled correlation.

Sanity Checks

Before presenting results, run these checks:

- Direction test: Does output move in the expected direction when you increase each input? If NPV rises when the discount rate rises, something is wrong.
- Magnitude test: Is the size of the swing economically plausible? A $\pm 1\%$ change in revenue should not swing NPV by 300%.
- Limiting case test: Set a key driver to zero or to an extreme value and confirm the model behaves correctly (e.g., zero revenue should produce a large negative NPV, not a positive one).
- Sum-of-parts check: Verify that the sensitivity contributions from individual inputs roughly account for the total variance you observe across scenarios.

Worked Example: A Flawed Attempt, Corrected

Setup. An analyst is valuing a three-year project. The base-case free cash flows are \$200 k, \$250 k, and \$300 k, and the discount rate is 10%. She wants to show how NPV changes if the discount rate moves to 8% or 12%, and if Year 3 cash flow changes by $\pm 20\%$.

Step 1 — Compute the base-case NPV (initial capital outlay = \$600 k).

$$\text{NPV} = -600 + 200/1.10 + 250/1.10^2 + 300/1.10^3$$

$$\text{NPV} = -600 + 181.82 + 206.61 + 225.39 = +\$13.82 \text{ k}$$

Step 2 — The flawed sensitivity table. The analyst builds a table varying both the discount rate and Year 3 cash flow simultaneously in every cell, labeling it a "sensitivity analysis." This is actually a two-variable scenario grid—fine as a tool, but she then reports the cell where rate = 8% and CF $f = +20\%$ as "the upside sensitivity to discount rate," ignoring that she has also changed cash flow.

Step 3 — Correct approach: isolate each variable.

Discount rate sensitivity only (Year 3 CF held at \$300 k):

| Rate | NPV |

|-----|-----|

| 8% | $-600 + 185.19 + 214.33 + 238.15 = +\37.67 k |

| 10% | \$13.82 k (base) |

| 12% | $-600 + 178.57 + 199.07 + 213.53 = -\8.83 k |

Year 3 CF sensitivity only (rate held at 10%):

| CF f | NPV |

|-----|-----|

| \$240 k (−20%) | $13.82 - (60/1.10^3) = 13.82 - 45.08 = -\31.26 k |

| \$300 k (base) | \$13.82 k |

| \$360 k (+20%) | $13.82 + 45.08 = +\$58.90 \text{ k}$ |

Step 4 — Sanity checks. NPV falls when the rate rises ' . NPV rises when CF f rises ' . The project flips from positive to negative NPV with only a 2-percentage-point rate increase—a meaningful flag for management that the deal has limited margin of safety.

Step 5 — Correct labeling. The analyst now presents two separate one-way tables and, separately, a clearly labeled two-variable grid for combined scenarios. Decision-makers can see the isolated effect of each driver before examining interactions.

The corrected analysis tells a richer, more honest story: the project is marginally positive at base, highly sensitive to the discount rate, and could easily turn negative under realistic adverse conditions—exactly the insight scenario planning is designed to surface.

